

GETTING STARTED WITH C/C++

& Introduction to Data Structures and Algorithms

11-Day Training Handout for 2nd Year Engineering Students

Task No.	Case Study	Business Problem Statement	Concepts Covered	Sample Input	Expected Output
1	Retail Billing System	A supermarket wants to automate customer billing. Develop a program to accept prices of three products, calculate the subtotal, GST (18%), and generate the final bill.	Variables, Input/Output, Arithmetic Operators, Formatted Output	Milk=60, Bread=45, Rice=250	Subtotal=355, GST=63.90, Final Bill=418.90
2	Employee Salary Processing	HR needs a utility to calculate an employee's monthly salary. Accept Basic Salary and calculate HRA (20%), DA (40%), and Net Salary.	Variables, Arithmetic Expressions	Basic Salary=30000	HRA=6000, DA=12000, Net Salary=48000
3	Student Performance Analyzer	A university wants to generate semester reports. Accept marks of five subjects and calculate Total, Average, and Percentage.	Variables, Arithmetic Operations	78,85,91,80,76	Total=410, Average=82, Percentage=82%
4	Currency Exchange System	A travel agency needs to convert Indian Rupees into US Dollars before issuing foreign exchange. Assume 1 USD = ₹86.	Variables, Formula Implementation	INR=17200	USD=200
5	Bank Interest Calculator	A bank wants to estimate the Simple Interest earned by customers based on Principal Amount, Rate of Interest, and Time Period.	Formula Implementation	P=50000, R=7, T=3	Interest=10500
6	Weather Monitoring System	A weather station stores temperature in Celsius but international reports require Fahrenheit. Develop a converter.	Arithmetic Expressions	Celsius=30	Fahrenheit=86.00
7	Package Delivery Cost Estimator	A courier company charges delivery based on package weight. Accept weight (kg) and cost per kg, then calculate the total delivery charge.	Variables, Multiplication	Weight=8 kg, Rate=45	Total Charge=360
8	Mobile Recharge Billing	A telecom operator wants to generate recharge invoices. Accept Recharge Amount and calculate GST (18%) and Final Amount Payable.	Arithmetic Operators	Recharge=499	GST=89.82, Total=588.82

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9	Character Encoding Utility	A software company is developing a keyboard utility. Accept a character and display its ASCII value for internal processing.	Character Data Type	Character='P'	ASCII Value=80
10	Restaurant POS (Point of Sale) System	A restaurant wants to automate billing. Accept quantities and prices of Pizza, Burger, and Cold Drink, calculate Subtotal, GST (18%), and Grand Total. Display a professional invoice.	Variables, Arithmetic, Formatted Output	Pizza=2×250, Burger=1×120, Drink=2×60	Invoice with Item Total, GST, Grand Total

Expected Deliverables

Submission Item	Description
Source Code	One .c file per problem
Program Output	Screenshot of successful execution
Algorithm	5–10 steps for each case study
Pseudocode	Before writing the actual code
Test Cases	At least 3 different input/output combinations

Placement Skills Developed

Skill	Industry Relevance
Problem Analysis	Requirement gathering
User Input Handling	Data collection from users
Business Logic Implementation	Core software development
Mathematical Computation	Finance, Retail, Banking, HR applications
Formatted Output	Invoice and report generation
Testing with Sample Data	Software testing fundamentals
Algorithm Thinking	Foundation for Data Structures and Coding Interviews